

Crypto Price Prediction

Rolling Backtesting · Baselines · LSTM · Ablation Study

Group Members:

Tae Moon

Ronnie Yin

Javier Ramos

Abhijit Rao

Hanish Kishan

Project Overview

01

Data

Yahoo Finance · yfinance
ETC-USD daily OHLCV
2017 – 2026, ~3 031 rows

02

Baselines

Last Value (tomorrow = today)
7-Day Moving Average
MAE · RMSE · Dir. Acc.

03

Lag + Ridge

30 return lags + log_volume
+ volatility_14 · Pipeline
Optuna alpha (32 features)

04

LSTM

64→32 units · dropout
Early stopping · ReduceLROnPlateau
Next-day return target

05

Ablation

4 groups: lags, +volume,
+volatility, all 32
Permutation importance + PDP/ICE

06

Backtest

Ridge +110% (Sharpe 1.04)
LSTM -46% · Buy&Hold -27%
ETH-USD bear market test

Dataset & Features

Source & Split

Provider: Yahoo Finance via yfinance

Asset: ETH-USD (Ethereum) — configurable

Range: 2017-01-01 to present

Rows: 3 031 daily records (OHLCV)

Preprocessing: Forward-fill, dropna

Timezone-naive datetime index

Split: Time-based only — no shuffle

Train 70% Val 15% Test 15%

(2 121 / 455 / 455 rows)

Derived Features

Returns r_t

$$(P_t - P_{t-1}) / (P_{t-1} + \epsilon)$$

Day-over-day return (stationarity-friendly)

volatility_14

14-day rolling std(returns)

Backward-looking only

log_volume

$$\log(1 + \text{volume})$$

Log-transformed volume

Return lags 1–30

$$r_{t-1}, r_{t-2}, \dots, r_{t-30}$$

30 return lags + 23 extra features (~53 total)

Evaluation Framework & Metrics

MAE

Mean Absolute Error

$$\text{mean } |y_i - \hat{y}_i|$$

Magnitude of price error

RMSE

Root Mean Squared Error

$$\sqrt{\text{mean}(y_i - \hat{y}_i)^2}$$

Penalizes large errors more

Dir. Acc.

Directional Accuracy

% correct up/down sign

Practical trading signal

Methodology Principles

- › Time-based splits only — no data shuffling to prevent future leakage
- › Expanding-window walk-forward: retrain Ridge on all past data per test window
- › Optuna (Bayesian) for Ridge alpha; objective: validation MAE on returns
- › Same test period for all models: baselines, Ridge, and LSTM
- › Ablation: six incremental feature groups (lags, volume/vol, technicals, multi-horizon, calendar, BTC)

Baseline Models

Baseline 1 — Last Value

$$P_{t+1} = P_t$$

Tomorrow's price equals today's price.

No information from volume or direction.

Dir. Acc. is not meaningful (predicts no change → often 0%).

0.558

MAE (ETH-USD)

0.848

RMSE (ETH-USD)

- ✓ Best on MAE & RMSE across all assets
- X Directional accuracy is not meaningful

Baseline 2 — 7-Day Moving Average

$$P_{t+1} = (1/7) \sum_{i=0}^6 P_{t-i}$$

Smoothed price over 7 prior days.

Captures local trend, filters single-day noise.

Dir. Acc. is meaningful (predicts a direction).

0.848

MAE

55.2%

Dir. Acc.

- ✓ Best directional accuracy among baselines (55.3%)
- X Higher MAE/RMSE than last value (more smoothing error)

Ridge Regression with Lag Features

Feature Engineering & Pipeline

Features (32 total): 30 return lags + log_volume + volatility_14

Scaler: StandardScaler on all 32 columns via ColumnTransformer

Model: Ridge regression (linear; regularized)

Hyperparameter search: Optuna (Bayesian) over Ridge alpha; objective: validation MAE on returns
Best alpha applied to test; permutation importance and PDP/ICE computed on test set

Target: Next-day return (stationarity-friendly); price via $P_{t+1} = P_t(1 + \hat{r}_t)$

Test Set Results (ETH-USD)

0.524

MAE

1.178

RMSE

52.0%

Directional Accuracy

Rolling Backtest Results

Train on past only → predict next day → roll forward
Retrain on all past data before each test window (expanding window)

Mean ± std of MAE and Dir.Acc across multiple test windows

Higher error than fixed split — rolling is stricter (retrain daily)

LSTM Architecture & Results

Architecture

Target: Next-day return r_t (not raw price)

Converted: $P_{t+1} = P_t \cdot (1 + \hat{r}_t)$

Sequence: Length 30; 3 features per timestep
(ret, log_volume, volatility_14)

Scaling: StandardScaler fit on train; applied to val/test

Layers: LSTM(64) $\rightarrow$ LSTM(32) $\rightarrow$ Dense(1) + Dropout(0.2)

Training: MSE loss $\cdot$ batch 32 $\cdot$ early stopping on val loss $\cdot$ ReduceLROnPlateau

Test Set Results (ETH-USD)

0.518

MAE

0.905

RMSE

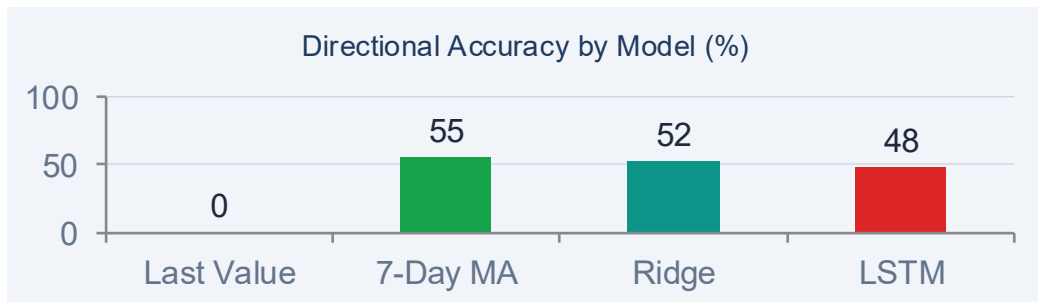
48.2%

Directional Accuracy

Key observation: LSTM is competitive on MAE (0.518) but directional accuracy (48.2%) is below Ridge. No consistent edge over simpler models across assets.

Full Model Comparison (ETH-USD, Test Set)

Model	MAE	RMSE	Dir. Acc.	Strength
Last Value	0.558 ★	0.848 ★	0% *	Best magnitude
7-Day MA	0.848	1.229	55.2% ★	Best direction
Ridge	0.524	—	~52%	Best ML dir.
LSTM	0.518	—	~48%	~50% dir.



* Last value predicts no change; directional accuracy = 0% is not meaningful.

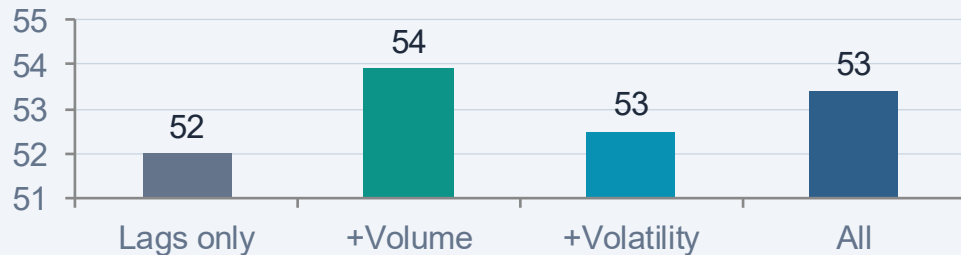
Key Takeaways

- › Last value wins MAE/RMSE — hard to beat
- › 7-day MA wins direction at 55.2%
- › Ridge is best ML model on direction (~52%); LSTM slightly below

Feature Ablation — What Actually Helps?

Feature Set	# Features	MAE	RMSE	Dir. Acc.
Lags only	30	0.806 ★	1.098 ★	52.0%
Lags + Volume	31	0.892	1.172	53.9% ★
Lags + Volatility	31	0.815	1.109	52.5%
All (Lags + Vol + Volat.)	32	0.895	1.177	53.4%

Directional Accuracy by Feature Set (%)



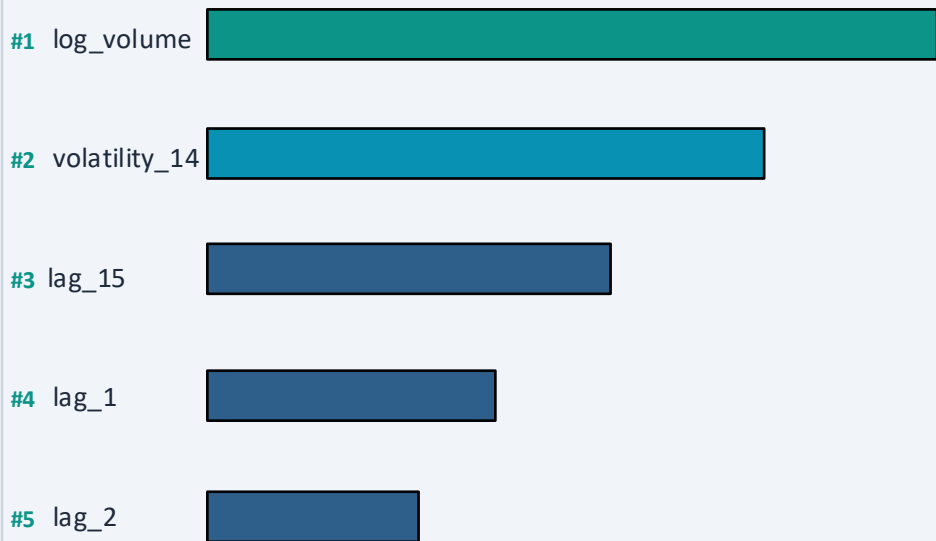
Ablation Insight

- › Lags only best for MAE / RMSE
- › Adding volume: largest direction gain (+1.9 pp vs lags only)
- › Volatility adds stability; all features = 32 total
- › +Volume gives the largest direction gain at small cost to magnitude

Permutation Importance · PDP / ICE

Permutation Importance (Best Lag+Ridge)

Shuffle each feature 10x · measure MAE increase · ETH-USD test set



Top-3 features are log_volume, volatility_14, and lag_15 (mid-range lag). Surprisingly, recent lag_1 is less important than volume and volatility.

PDP / ICE Insights

lag_1

Roughly linear PDP — higher lag_1 → higher prediction. Effect is homogeneous across ICE curves (Ridge is linear in features).

log_volume

Flatter PDP — volume has a weaker marginal effect on price than lag features, but matters for directional accuracy (ablation).

ICE curves for lag_1 show similar slope across samples — the linear effect is consistent, not sample-specific.

Where Does the Model Fail?

Long-Tailed Residuals

Residual histogram is centered near zero but has heavy tails — large errors occur in both directions, indicating the model cannot handle sharp price moves.

Clustered Errors Over Time

Errors are not uniformly distributed across time. They cluster in volatility regimes — calm periods have smaller errors; turbulent periods have large errors.

High-Volatility Failure

$|\text{Residual}|$ vs volatility_{14} plot shows a positive correlation — larger absolute errors occur when the 14-day rolling volatility is high. Model performs worst in volatile markets.

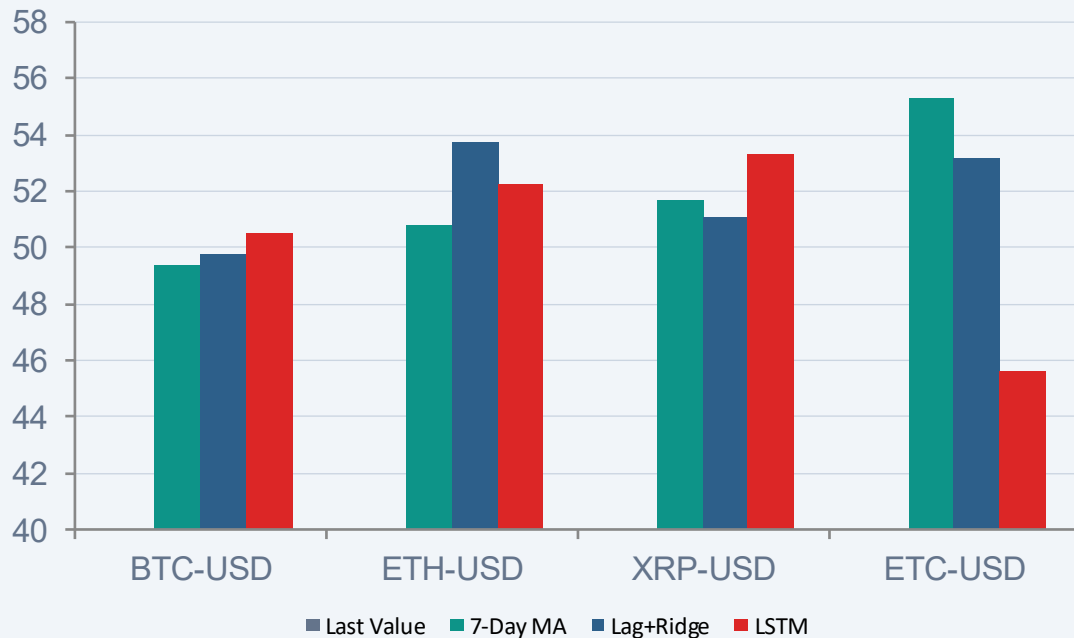
Smoothed Extrapolation

Lag+Ridge behaves like a smoothed extrapolation of recent price lags. It systematically under- or over-predicts during sharp moves and mean-reverting spikes.

Conclusion: Lag+Ridge (and simpler models) are smoothing extrapolators — they fail most in high-volatility regimes and on large price moves.

Generalization Across Cryptocurrencies

Directional Accuracy by Asset & Model (%)



BTC-USD

Best dir.: LSTM dir. edge (50.5%)

All models near 50%; last value best MAE

ETH-USD

Best dir.: Lag+Ridge (53.7%)

Slightly more directionally predictable

XRP-USD

Best dir.: LSTM (53.3%)

Only asset where LSTM wins direction

ETH-USD

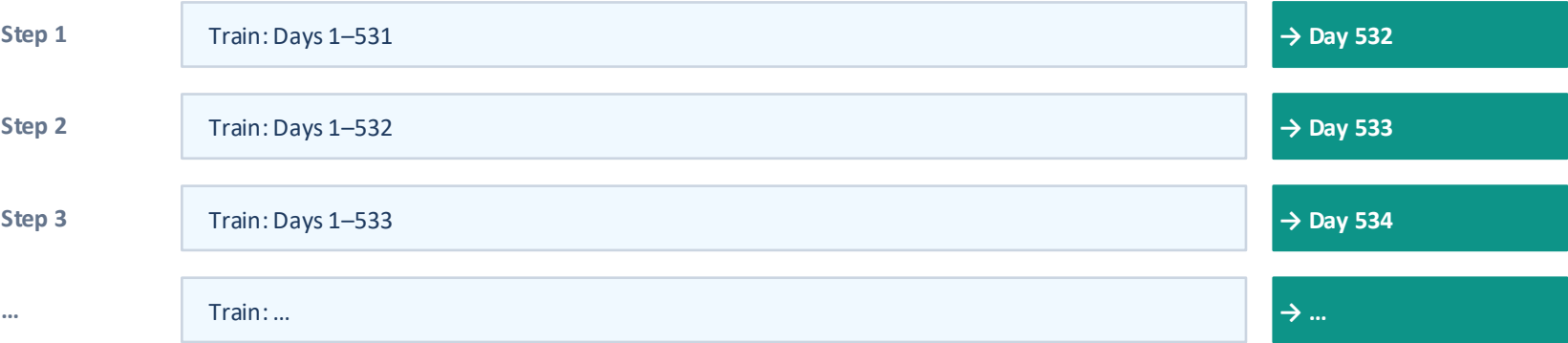
Best dir.: 7-Day MA (55.3%)

Best directional result; LSTM < 50%

Last value wins MAE/RMSE on every asset. ETH-USD results shown; 7-day MA leads on direction (~55%); Ridge is best ML model on direction.

Expanding-Window Walk-Forward — Ridge

How Expanding-Window Evaluation Works



Minimum training window required; retrain Ridge before each test window

Fixed Split vs. Expanding-Window — Ridge (ETH-USD)		
Evaluation	Fixed-Split Test	Expanding-Window
MAE	0.895	1.38
RMSE	1.178	3.49
Dir. Acc.	53.2%	52.7%

Key Findings

Baselines are competitive: Last value achieves best MAE/RMSE across all assets — a surprisingly strong benchmark.

Ridge on direction: Best ML model — ~52% Dir.Acc on ETH-USD test set.

LSTM: LSTM Dir.Acc ~48%; comparable MAE/RMSE to Ridge on price.

Ablation (6 groups): Volume improves direction most (+1.9 pp). Full feature set (~53) best for direction; lags-only best for MAE/RMSE.

Error analysis: All models fail in high-volatility regimes and on large moves. Expanding-window is stricter than fixed-split.